

Capstone Final Project



Spotify Recommendation Algorithm

Explaining and Predicting Spotify Recommendations
Using Audio Features and Listener Behavior

STAT - 6399

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Introduction

Music streaming platforms such as Spotify have dramatically changed the way people listen to music. Instead of purchasing albums or relying on radio stations, listeners now have instant access to millions of songs from a wide variety of artists and genres. Because of this massive availability of music, streaming services rely heavily on data analysis and machine learning techniques to organize songs and recommend music that users may enjoy. Understanding patterns within music data has therefore become an important area of study in both data science and the music industry.

Spotify provides many measurable audio features that describe different musical characteristics of a song. These features include variables such as danceability, energy, acousticness (amount of acoustic content), loudness (intensity of sound), speechiness (presence of spoken words), instrumentalness (number of instruments used), valence, and tempo. Each of these variables captures a specific aspect of how a song sounds. For example, danceability measures how suitable a song is for dancing, energy reflects the intensity of the music, and acousticness indicates whether the track relies more on acoustic instruments rather than electronic production. By analyzing these features together, it becomes possible to better understand how songs differ from one another and whether certain types of music share similar characteristics.

The purpose of this project is to apply data analytics and machine learning techniques to analyze Spotify music data and uncover meaningful patterns within the dataset. The analysis includes exploratory data analysis (EDA), data preprocessing and feature preparation to understand the distribution of musical features and identify relationships between variables. Exploring the data helps provide initial insights into how different audio features interact and how songs vary across multiple musical dimensions. Since the dataset contains variables measured on different scales, appropriate data preparation techniques such as feature selection and variable scaling are applied. These steps ensure that the data is suitable for statistical analysis and machine learning algorithms.

One of the main analytical methods used in this study is unsupervised machine learning, specifically clustering techniques. Clustering allows observations to be grouped together based on similarities in their characteristics without requiring predefined labels. In this project, clustering is used to identify natural segments of songs based on their audio features. Techniques such as the Elbow Method and Silhouette Analysis are used to determine the optimal number of clusters before applying the K-means clustering algorithm.

Overall, this project demonstrates how data analytics, statistical techniques, and machine learning methods can be applied to large music datasets to uncover patterns and gain insights into musical structure. By analyzing Spotify audio features and identifying clusters of similar songs, this study provides a data-driven approach to understanding music characteristics and segmentation within modern streaming platforms.

Literature Review

The rapid growth of digital music streaming services has increased the significance of managing large music libraries and comprehending musical preferences. Listeners find it difficult to manually peruse every song on services like Spotify, which provide millions of songs to users worldwide. Machine learning and data mining techniques are now essential tools for analyzing music data and enabling recommendation systems. Researchers are increasingly utilizing clustering algorithms and aural characteristics to find patterns in music in order to better categorize and suggest songs.

One of the most important aspects of modern music analysis is the use of audio features extracted from songs. Spotify provides measurable characteristics such as danceability, energy, acousticness, loudness, speed, and valence that describe a song's sound. These characteristics allow scholars to use numerical data to examine music rather than relying just on genre classifications or subjective descriptions. These auditory features can be used to assess musical patterns, understand listener preferences, and even predict the popularity or mood of a song, according to research (Saragih, 2023). Furthermore, researchers have demonstrated that analyzing combinations of auditory features might reveal deeper relationships between musical attributes and listener behavior.

Numerous studies have examined the use of machine learning algorithms to analyze Spotify datasets. A common technique that groups music with similar characteristics without the requirement for labeling is clustering. Clustering is particularly useful for music analysis because musical styles often overlap and are not always clearly separated by traditional genre designations. Li (2024) claims that clustering methods such as K-means clustering may group songs based on similarities in auditory qualities, allowing researchers to identify musical passages and identify trends in large datasets. This method is often used in conjunction with Principal Component Analysis (PCA) and other dimensionality reduction approaches to reduce complex datasets while preserving the most important information.

K-means clustering has been widely used in music research due to its simplicity and effectiveness when working with numerical data. The approach groups observations by reducing the distance between songs and cluster centers, allowing similar songs to be clustered together. Previous research has shown that musical patterns, such as clusters indicating instrumental, acoustic, or energetic dance music, can be found by applying K-means clustering to Spotify audio data. For example, Duman et al. (2022) found that songs could be separated into several important subgroups using K-means clustering on Spotify audio data, including "happy-energy," "sad-instrumental," and "soft-acoustic" clusters. Their findings demonstrated that measurable musical traits can naturally form structured groupings even in the absence of predefined categories.

In addition to clustering, many studies explore how machine learning techniques could improve music recommendation systems. Recommendation systems aim to suggest songs that users might enjoy based on trends in listening habits or aural similarities. Music recommendation systems are built on algorithms that analyze large datasets and identify relationships between songs, genres, and listening habits (Gong et al., 2020). Clustering is essential to this process because it helps organize songs into groups of related tracks, which can then be used to recommend appropriate music to listeners.

Dalida et al. (2023) used machine learning approaches to investigate the connection between Spotify audio attributes and emotional reactions to music. According to research, factors like intensity, pace, and valence can have a significant impact on how listeners interpret a song's mood. These results imply that auditory characteristics offer significant insights into musical qualities and can contribute to the explanation of why particular songs are appealing to listeners.

Finding trends and patterns in Spotify music data is the subject of another field of study. Marlia (2024) investigated the relationship between song popularity on Spotify and attributes including danceability, intensity, and loudness. The study discovered that songs could be categorized into distinct groups that represented high-energy, medium-energy, and low-energy musical styles by using clustering techniques and feature value analysis. Researchers can gain a better understanding of how musical qualities affect listener engagement and popularity by using these kinds of analyses.

Previous research indicates that Spotify's audio features provide valuable data for analyzing musical trends and audience preferences. Researchers can use machine learning techniques like dimensionality reduction and clustering to uncover hidden structures in large music collections. These methods make it possible to classify songs into pertinent groups and make it easier to develop recommendation systems that help consumers find music more efficiently. The current study builds on earlier research by using clustering algorithms on a large Spotify dataset to examine whether songs naturally produce distinct musical portions depending on their acoustic attributes.

Dataset Description

The dataset used in this project consists of Spotify songs and their associated audio features. It was obtained from a publicly available Spotify dataset that contains detailed information about individual tracks and their musical characteristics. The dataset includes approximately 114,000 songs and 21 variables, providing a large sample of music across different genres and styles. Each observation in the dataset represents a single song along with measurable features that describe how the song sounds.

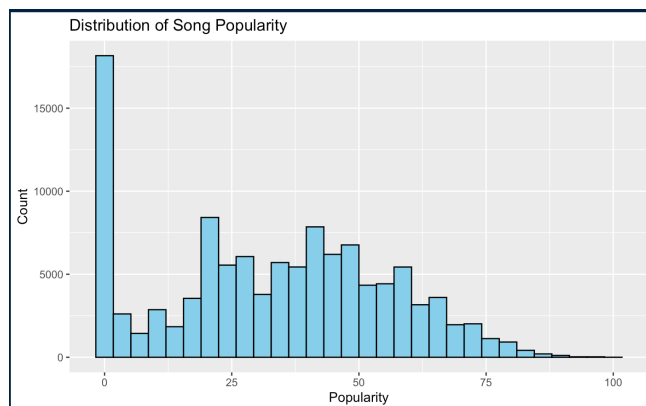
The dataset contains both descriptive and numerical variables. Descriptive variables include information such as the track name, artist name, track ID, and genre label, which help identify the songs but are not useful for mathematical clustering analysis. In addition to these descriptive variables, the dataset includes a variety of audio features generated by Spotify's music analysis system. These numerical features capture different aspects of the musical structure of each song.

Before performing clustering analysis, the dataset was prepared by selecting the relevant numerical audio features and removing non-numeric identifiers such as track names and artist names. Since clustering algorithms rely on numerical distances between observations, only variables that represent measurable musical characteristics were included in the analysis. The explicit content variable, which originally appeared as a TRUE/FALSE value, was converted into a binary numerical variable to make it compatible with the clustering process.

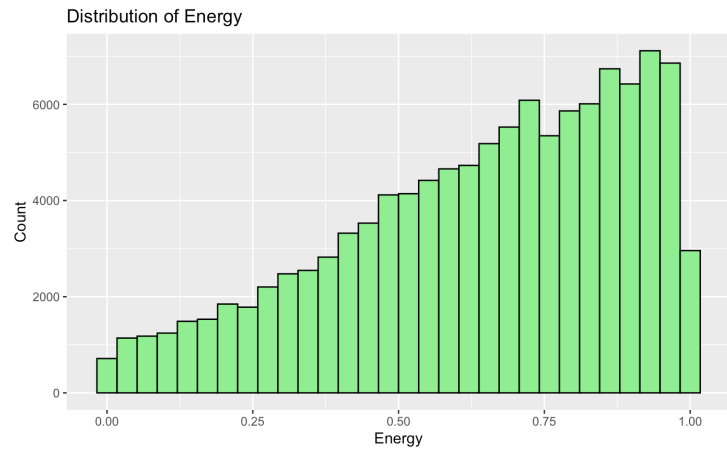
In addition, the selected features were standardized using a scaling procedure. This step was necessary because the variables in the dataset are measured on different scales. For example, tempo is measured in beats per minute, while danceability and energy are values between 0 and 1. Standardizing the variables ensures that no single feature dominates the clustering results simply because of its numerical range.

By focusing on these audio characteristics, the dataset provides a strong foundation for identifying patterns and similarities among songs. This allows the clustering algorithm to group songs with similar musical properties and uncover natural segments within the Spotify music library.

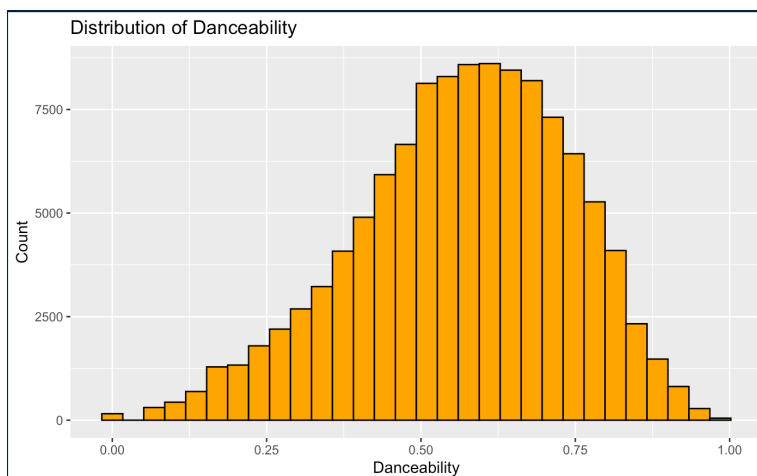
Exploratory Data Analysis



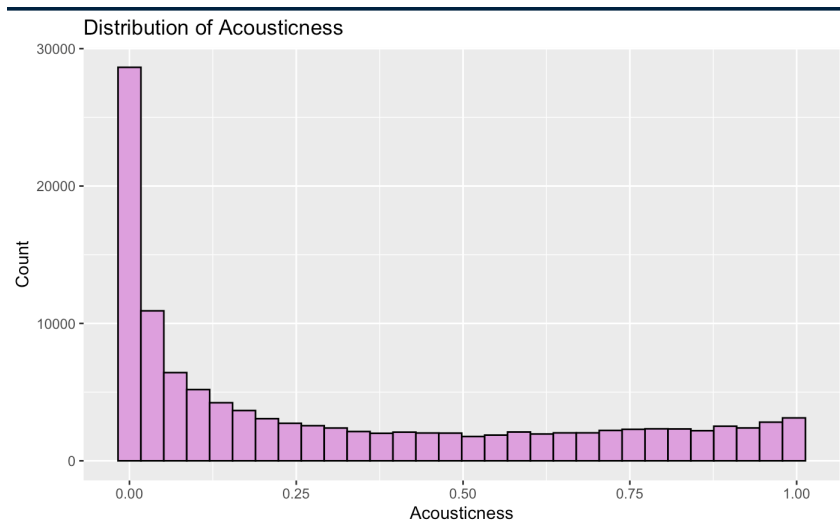
This graph shows how popularity is distributed across songs in the dataset. If the distribution is skewed, it suggests that most songs have lower to moderate popularity, while only a smaller number of tracks are highly popular.



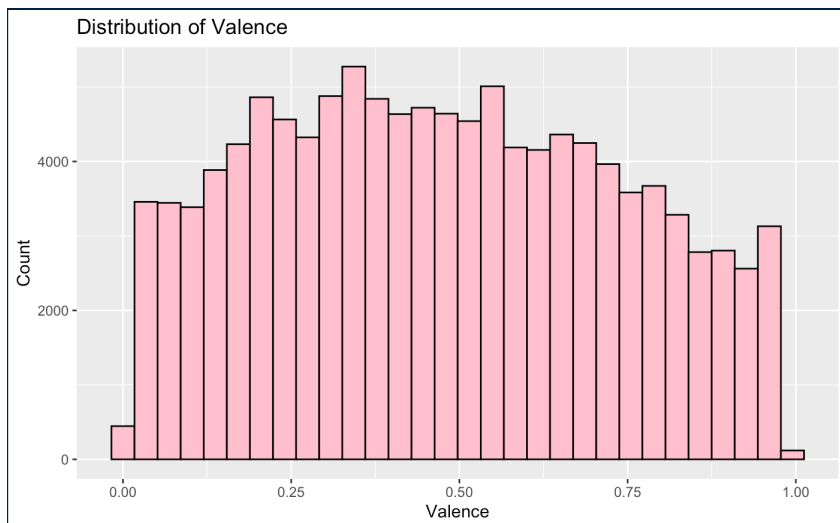
The distribution of energy helps show whether the dataset contains more calm songs or more intense songs. This is important because energy later becomes one of the key variables that separates clusters.



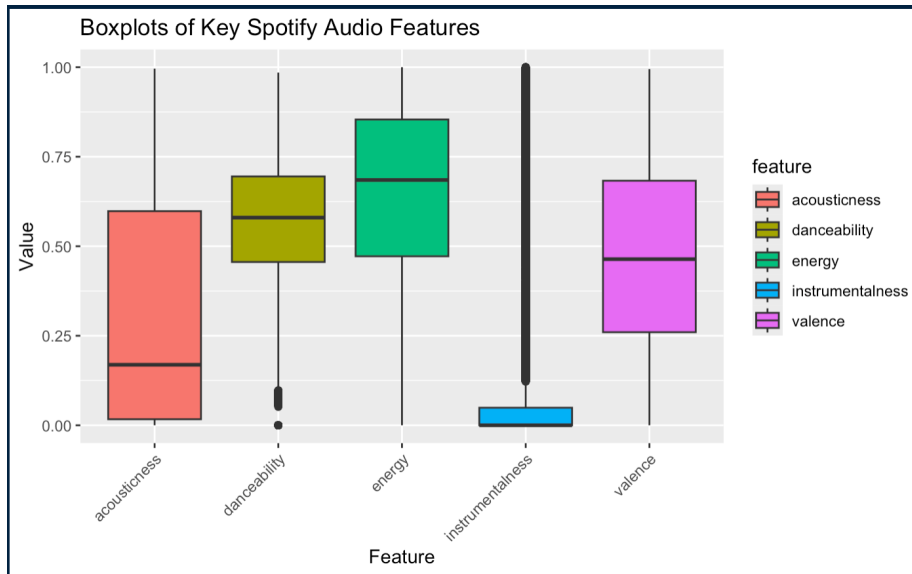
Danceability indicates how suitable a song is for dancing. This distribution provides an early view of whether the dataset is dominated by highly danceable songs or contains a wider range of musical styles.



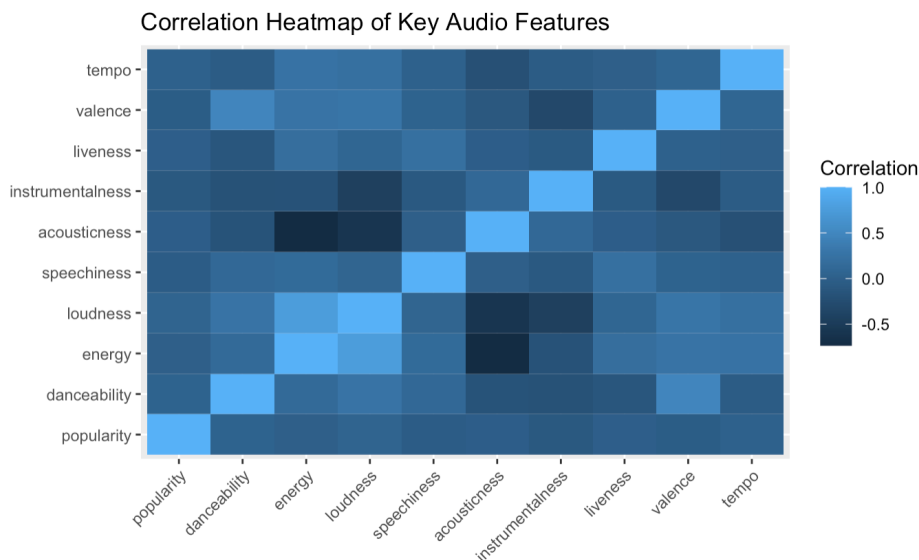
Acousticness measures the extent to which a song relies on acoustic instrumentation. This feature is useful for identifying whether the dataset contains more electronically produced music or more acoustic tracks.



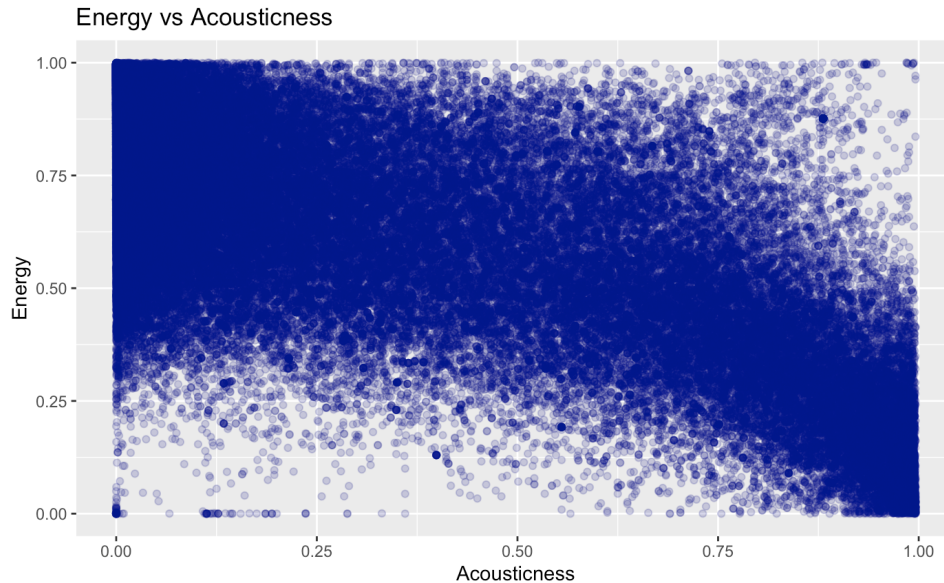
Valence reflects the emotional tone of a song, ranging from more negative or sad to more positive or happy. Examining valence provides insight into the emotional diversity of the music in the dataset.



The boxplots show that the audio features vary in both central tendency and spread, indicating that the dataset includes songs with diverse musical characteristics. This variability supports the use of clustering, since clustering works best when observations differ across multiple dimensions.



The correlation heatmap provides an overview of the relationships among key audio features. Strong positive or negative correlations suggest that some musical characteristics tend to occur together. For example, energetic songs may also be louder, while acoustic songs may show lower energy levels. These relationships help explain why certain clusters later emerge in the segmentation analysis.



This scatterplot helps show the contrast between acoustic and energetic songs. A negative relationship would suggest that songs with stronger acoustic qualities tend to have lower energy, which aligns with the soft and instrumental clusters identified later.

Before performing clustering analysis, exploratory data analysis was conducted to better understand the structure and characteristics of the Spotify dataset. The dataset contains over 100,000 songs along with multiple audio features such as danceability, energy, acousticness, instrumentalness, loudness, valence, and tempo. Summary statistics and missing value checks were first examined to ensure that the data was suitable for analysis.

Several visualizations were created to study the distributions of important variables. Histograms of popularity, energy, danceability, acousticness, and valence showed that the songs vary considerably across musical characteristics, indicating that the dataset contains a wide range of song types. Boxplots of the key audio features further confirmed that the variables differ in spread and central tendency, which supports the use of clustering methods for identifying song segments.

A correlation heatmap was used to examine relationships between variables. The results suggested meaningful associations among several audio features. For example, energy and loudness appeared to be positively related, while acousticness showed a negative relationship with energy. Scatterplots such as energy versus acousticness and danceability versus valence provided further visual support for these patterns. These relationships suggest that the audio features contain structure and variation, making them useful for segmentation analysis.

Overall, the exploratory data analysis showed that the dataset is diverse, well suited for clustering, and contains meaningful relationships among the musical features. These findings provided the foundation for the next stage of the project, which involved identifying distinct song segments using K-means clustering.

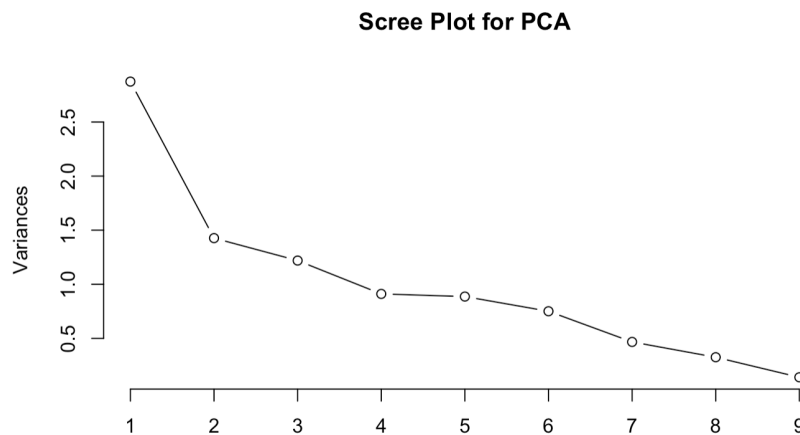
PCA

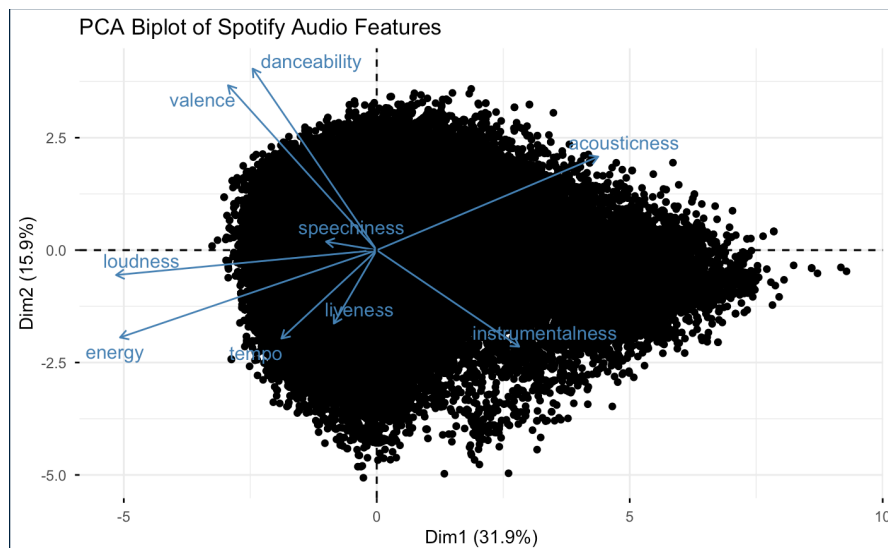
Principal Component Analysis (PCA) was performed to better understand the underlying structure of the Spotify audio features before applying clustering methods. Since the dataset contains several numerical variables that may be correlated with one another, PCA was used as a dimensionality reduction technique to identify the main directions of variation in the data. This helps simplify the dataset while preserving as much information as possible.

The PCA was conducted using selected numerical audio features including danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, and tempo.

The results of the PCA show how much variance in the dataset is explained by each principal component. The scree plot was used to examine the importance of the components and identify how many of them explain most of the variation in the audio features. The first few components accounted for the largest proportion of the variance, suggesting that the structure of the dataset can be summarized using a smaller number of dimensions.

The variable loadings and PCA biplot were then examined to interpret the meaning of the principal components. Features such as energy, loudness, and danceability may contribute strongly to one component, representing a dimension related to musical intensity or activity. Other features such as acousticness and instrumentalness may contribute to another component, representing a more acoustic or instrumental style of music. These results provide insight into the major patterns within the dataset and support the later clustering analysis by showing that songs vary across a few meaningful musical dimensions.





PCA Results

Principal Component Analysis (PCA) was conducted to examine the underlying structure of the Spotify audio features and identify the main dimensions that explain variation among songs. The analysis was performed using standardized audio features including danceability, energy, loudness, speechiness, acousticness, instrumentalness, liveness, valence, and tempo.

The results show that the first principal component (PC1) explains 31.9% of the total variance, while the second component (PC2) explains 15.9%. Together, the first two components account for 47.8% of the total variation in the dataset. When the third component is included, the cumulative variance explained increases to 61.3%. This indicates that a relatively small number of components capture a large portion of the variability in the musical features.

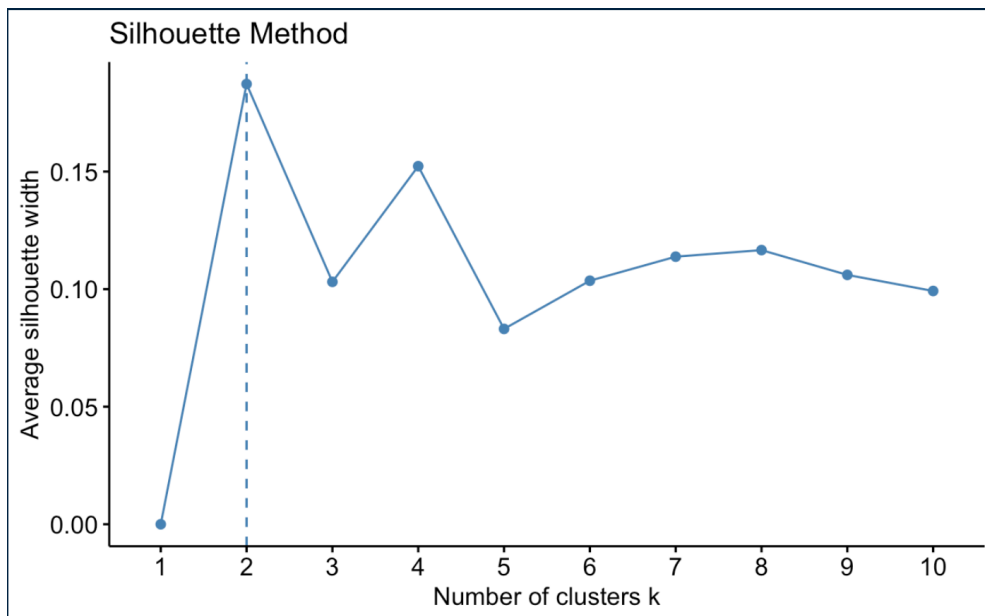
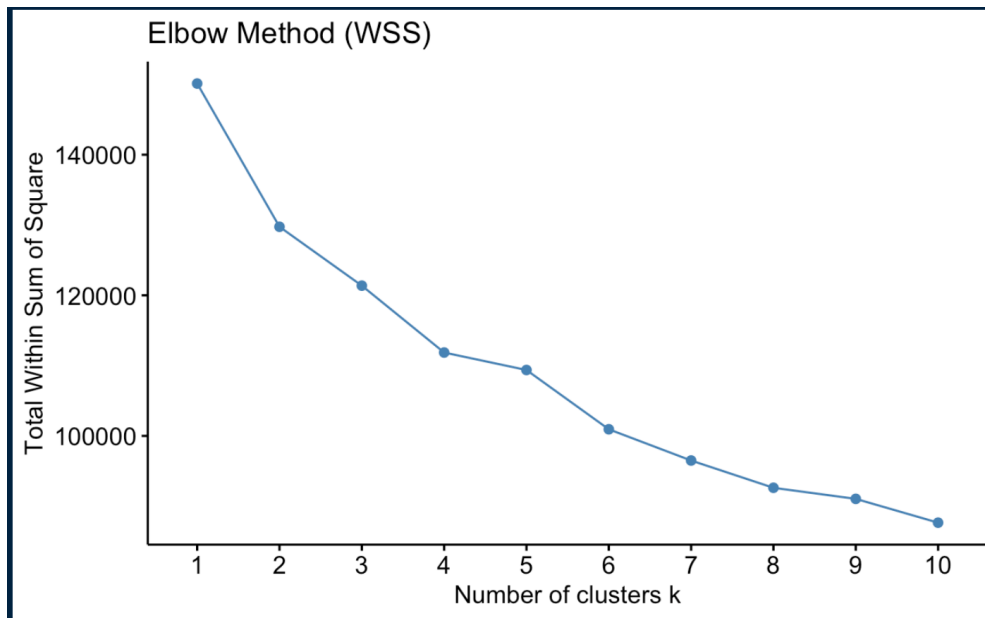
The PCA biplot provides insight into the relationships between the audio features. Features such as energy, loudness, and tempo appear closely aligned, suggesting that songs with higher energy levels also tend to be louder and faster. In contrast, acousticness and instrumentalness form a separate grouping, indicating that acoustic songs often contain more instrumental characteristics and less electronic production.

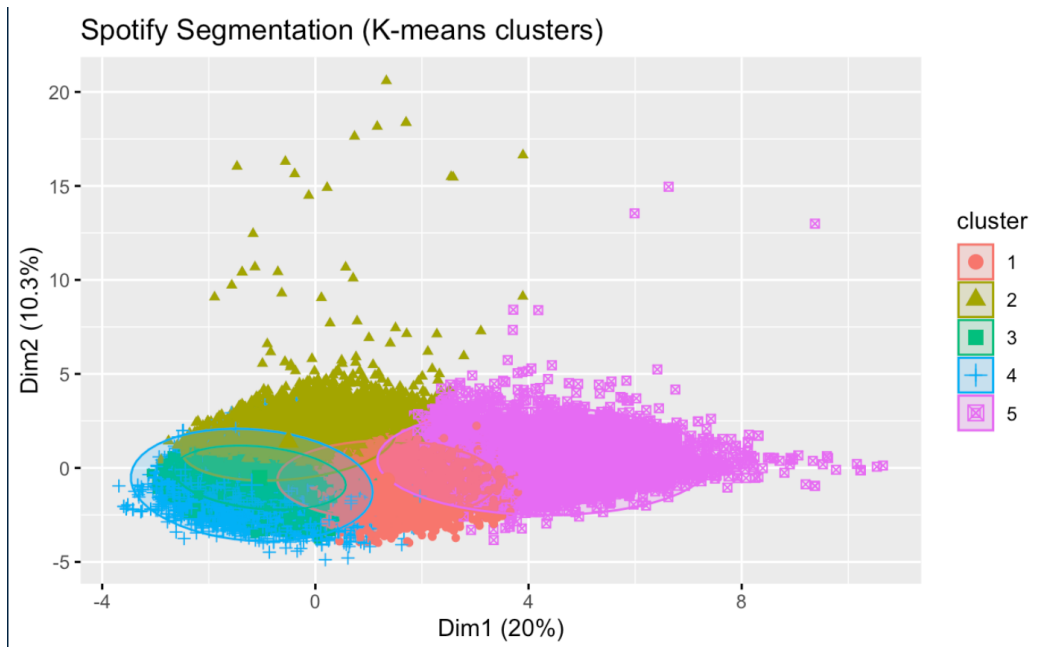
Another noticeable relationship is between danceability and valence, which suggests that songs with a more positive emotional tone tend to be more danceable. These patterns highlight several key musical dimensions, including intensity, mood, and acoustic qualities, which contribute to the variation among songs in the dataset.

Overall, the PCA results demonstrate that the Spotify audio features contain meaningful structure. The identified components reflect distinct musical characteristics that help explain how songs

differ across multiple dimensions. These findings support the subsequent clustering analysis, as the presence of clear feature relationships suggests that songs can naturally form groups based on their audio properties.

Clustering Analysis





- Cluster 1 = Acoustic & Soft Listeners
- Cluster 2 = High-Intensity / Workout Listeners
- Cluster 3 = Upbeat Pop / Party Listeners (largest cluster)
- Cluster 4 = Rap / Hip-Hop Focused Listeners
- Cluster 5 = Instrumental / Study & Ambient Listeners

Results

To identify distinct music preference segments, K-means clustering was applied to standardized Spotify audio features. The number of clusters was determined using both the Elbow Method and Silhouette analysis, which indicated that a five-cluster solution provided a meaningful balance between model simplicity and cluster separation.

The clustering algorithm grouped songs based on similarity across multiple dimensions including danceability, energy, acousticness, instrumentality, loudness, tempo, and valence. Since the analysis was unsupervised, no prior labels were used; instead, the algorithm identified natural groupings within the data.

The five clusters revealed clearly distinguishable musical profiles:

Cluster 1 is characterized by high acousticness and lower energy levels, representing softer and more acoustic-oriented music. Songs in this group tend to be softer, less intense, and more acoustic in nature. These songs are likely linked to calm listening environment such as relaxing background music.

Cluster 2 exhibits very high energy and tempo with low acousticness, indicating high-intensity or workout-style music. Songs are loud, fast paced and intense, which are very suitable for high energy activities. This group represents music that is designed to motivate and energize listeners.

Cluster 3 shows high danceability, high energy, and high valence, consistent with upbeat and mainstream pop music. It likely represents mainstream music played at party and social settings. This is the largest cluster, and many songs fall under this category.

Cluster 4 is distinguished by elevated speechiness and strong rhythmic structure, suggesting rap and hip-hop influence. This includes songs that are rhythmic and follow a pattern and reflect characteristics associated with hip hop and rap music.

Cluster 5 demonstrates very high acousticness and instrumentality with low energy and loudness, representing instrumental, ambient, or study-oriented music. This includes tracks with instrumentality and ambient music and suitable for focusing. The clusters differ meaningfully across multiple features, indicating that Spotify songs naturally form distinct stylistic segments.

Null Hypothesis (H_0):

Spotify audio features do not form statistically meaningful or distinct clusters. Any grouping observed is due to random variation in the data.

Alternative Hypothesis (H_1):

Spotify audio features form statistically meaningful and distinguishable clusters representing distinct music preference segments.

The clustering results provide strong evidence against the null hypothesis that Spotify audio features do not form distinguishable groups. Instead, the findings suggest that music can be segmented into coherent preference categories based solely on measurable audio characteristics.

In practicality, these results demonstrate how streaming platforms can utilize multivariate feature analysis to segment music libraries and potentially align content with listener behavior. For example, high-energy clusters could support workout playlist generation, while instrumental clusters could be leveraged for focus or study playlists.

The segmentation also highlights the importance of multidimensional modeling. No single feature defined the clusters independently; it was the combination of energy, danceability, acousticness, and instrumentality that created meaningful separation.

Statistical Validation of Clusters

The following methods help evaluate whether the clusters differ significantly across key audio features and whether the grouping structure is well-defined.

Between-Cluster Variance Comparison

The K-means algorithm minimizes within-cluster variation by design. The results showed a clear reduction in within-cluster sum of squares as the number of clusters increased, with an optimal balance achieved at five clusters. This indicates that the clustering structure captures meaningful differences in the data.

ANOVA Across Clusters

ANOVA tests the null hypothesis that the means of a variable are equal across all clusters. If the test is significant, it suggests that at least one cluster differs from the others for that feature.

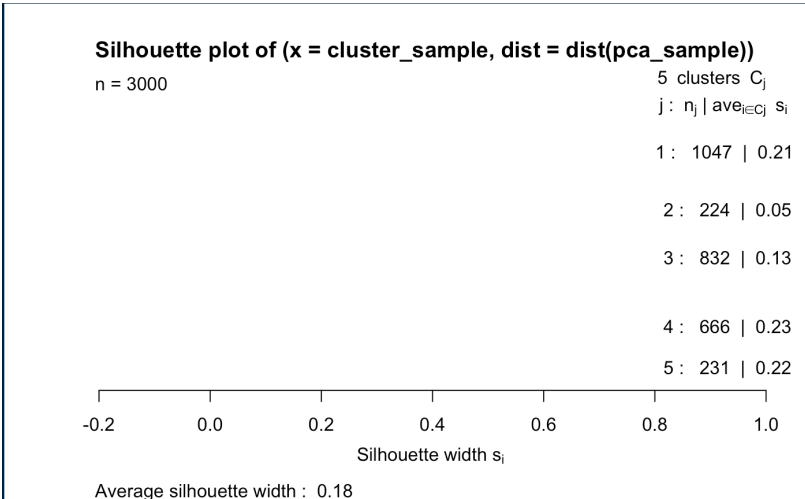
The results showed that several key variables, including energy, danceability, acousticness, and instrumentality, differed significantly across clusters. This confirms that the clusters are not arbitrary but are driven by meaningful differences in musical characteristics.

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
factor(cluster)	4	4611	1153	50511	<2e-16 ***
Residuals	113995	2601	0		

Signif. codes:	0 '***'	0.001 '**'	0.01 '*'	0.05 '.'	0.1 ' ' 1
	Df	Sum Sq	Mean Sq	F value	Pr(>F)
factor(cluster)	4	7827	1957	46692	<2e-16 ***
Residuals	113995	4778	0		

Signif. codes:	0 '***'	0.001 '**'	0.01 '*'	0.05 '.'	0.1 ' ' 1

Silhouette Score Analysis



The silhouette analysis was used to evaluate how well the clusters are separated. The average silhouette score obtained was **0.18**, which indicates that the clusters have **moderate separation**.

While the clusters are distinguishable, there is some overlap between them, suggesting that certain songs share similar characteristics across groups.

From the silhouette plot, some clusters show higher average values (around 0.21–0.23), indicating better-defined groupings, while others (such as cluster 2 with a lower score) are less clearly separated. This is expected in music data, where songs often share overlapping features such as tempo, energy, and mood.

Overall, the silhouette results suggest that the clustering structure is **reasonable but not perfectly distinct**, which aligns with the complex and continuous nature of musical characteristics.

Results and Interpretation

The analysis revealed clear patterns in how Spotify songs can be grouped based on their audio features. The clustering results identified distinct segments of songs characterized by differences in energy, danceability, acousticness, and valence. For example, some clusters represented high-energy and danceable tracks, while others captured more acoustic and low-tempo music. These groupings highlight how songs naturally organize into meaningful categories based on their sound characteristics.

The PCA results supported these findings by showing that much of the variation in the dataset could be explained using a smaller number of components, making it easier to visualize differences between clusters. Additionally, statistical validation using ANOVA confirmed that key features such as energy and acousticness significantly differ across clusters, while the silhouette analysis indicated moderate but reasonable cluster separation. Overall, the results suggest that Spotify songs can be effectively segmented using audio features, although some overlap between groups exists due to the continuous nature of music.

Limitations

Although the analysis provides useful insights into musical patterns within the Spotify dataset, several limitations should be considered. First, the study relied solely on song-level audio features and did not include user listening behavior or demographic information. While audio features capture many important musical characteristics, listener preferences may also be influenced by contextual factors such as personal taste, cultural background, and listening habits. Incorporating user-level data could provide a deeper understanding of how these clusters relate to real listener behavior.

Another limitation is that the clustering analysis depends on the selection of variables included in the model. Although the study used several important Spotify audio features, other variables such as lyrical content, genre metadata, or artist information were not included in the clustering process. Including additional features could potentially reveal more nuanced musical segments.

Finally, the clustering method used in this study was limited to the K-means algorithm. While K-means is widely used and computationally efficient, it assumes that clusters are relatively spherical and evenly distributed. Musical data may contain more complex structures that could be captured by other clustering techniques such as hierarchical clustering or density-based clustering. Future research could explore alternative clustering approaches to compare results and further refine the segmentation of music datasets.

Conclusion

This study explored whether Spotify audio features can be used to identify meaningful patterns and segments within a large collection of songs. By applying exploratory data analysis, principal component analysis, and K-means clustering, the project demonstrated that songs in the dataset vary across several key musical characteristics such as energy, danceability, acousticness, and emotional tone. The PCA results showed that a large portion of the variation in the dataset could be explained by a small number of underlying dimensions related to musical intensity, mood, and acoustic qualities. These findings confirmed that Spotify audio features contain meaningful structure that can be analyzed using statistical and machine learning techniques.

The clustering analysis further revealed that songs naturally form distinct groups based on their audio properties. Five clusters were identified, representing different musical profiles such as acoustic and soft music, high-energy workout songs, upbeat pop music, rap and hip-hop influenced tracks, and instrumental or ambient music. These results support the idea that measurable audio features can effectively capture differences in musical style without relying on traditional genre labels. Overall, the project demonstrates how data analytics and machine learning can be applied to large music datasets to better understand musical structure and support modern music recommendation systems used by streaming platforms like Spotify.

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